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Environmental Pollution

33.1. INTRODUCTION

- Environment refers to the ecological, economic,political, social and technology consideration that impact on engineering and in turn are impacted by the results of engineering.
- Scientists and engineers develop technology that impacts the environment in which we live.
- Sometimes back in a certain wooded area located remote from industry, the air was pure and the waters were clean.

  As the first few people settled in that area, the water had no problem in assimilating their wastes and the smoke from their fires could easily dissipate in the atmosphere.

  However, with the growth of population, industry thrived and the affluence of the community increased steadily. An almost unavoidable concomitant of this developmental pattern was the increase in water and the air pollution.
- Deterimental effects of increased waste, and water and air pollution are controlled through the implementation of environmental specifications and standards, which in turn place more stringent constraints on future technology.

33.2. ECOLOGY

- In general *Ecology* pertains to the study of relationships between various organisms and their environment. This includes consideration of plant, animal, and human populations in terms of rate of population growth, food habits, living habits, reproductive habits and ultimate death.
- Human ecology is a social science that studies the relationships between man and his environment.
- The ecological study of man may be divided into two fields :

(i) Human ecology proper, which studies the relationships between human biological factors and the natural environment

(ii) Social ecology, which studies the relationships among natural environment, population, technology and society.

- During last few decades, the world population growth combined with the technological changes associated with our living standards, has created a greater consumption of our resources; concurrently the amount of pollution waste has increased significantly. The net effects of this have caused alterations to the basic biological process and to some extent these alterations have been harmful. Of particular concern are those problems, dealing with air pollution, water pollution etc.

33.3. FACTORS CAUSING/AFFECTING POLLUTION

- The level of pollution is a function of so many factors. It varies directly with the level of population and per capita income and inversely with the extent of recycling, technology and waste treatment.

  P=*f* (Proplation, per capita income, 1/recycling, 1/technology, 1/treatment)

- Pollution increases with the population density, because, as the population increases, more burden is placed on the environment.
- Environmental pollution further increases as the standard of living goes up. This is becaue of the fact that, the goods and services demanded per person increase.
- Pollution is inversely related to the degree of recycling. In other words if the waste product is cleaned and reused, the pollution level will decrease.
- Technology also acts to ameliorate pollution. Engines which are more efficient produce less pollution/waste for the same degree of utility.
- The intensity of pollution also depends upon the amount of waste treatment. There are several methods available to clean air and water discharged from industrial processes.

For example, pollutants in air can be removed by electrostatic precipitators and waste waters can be cleaned by mechanical filtration to chemical and biological processing.

33.4. EFFECTS OF POLLUTION ON HUMAN HEALTH

- Pollution develops respiratory diseases which may cause disability and death.
- Given below is a list of various health problems associated with pollution.

(i) Occupational skin diseases e.g. chemical burns or inflammations.

(ii) Dust diseases of lungs e.g. silicosis found in foundry workers.

(iii) Respiratory conditions due to Toxic agents e.g. acute congestion due to chemicals, dusts, gases or fumes.

(iv) Poisoning e.g. by Pb, Hg,Cd, H_2S, benzol,plastics and resins etc.

(v) Disorders due to physical Agents such as environmental heat or low temperature.

(vi) Disorders due to repeated Trauma e.g. noise-induced hearing loss.

33.5. AIR POLLUTION AND CONTROL

Introduction

- Air pollution may be defined as any gaseous, liquid or solid material suspended in the air which creates an undesirable effect. Air pollution is injurious both to life and property.
- Polluted air is present over almost every industrialized community. Much of the blame for the contaminated atmosphere can be directed toward automotive exhaust but industrial wastes are also big contributors. Factories and electric-power generating plants emit sulphur dioxide and fly ash and microscopic pieces of metal or metal oxides.
- Air pollution is growing at a rate that alarms experts throughout the world and it will continue to grow despite measures being taken to alleviate it. Only the most drastic steps can reverse the present trend.

Sources of air pollution

- Sources of air pollution include
  (i) Factory chimneys
  (ii) Home furnaces
  (iii) Burning refuse
  (iv) Burning fuel for light, heat, power and transportation.
  (v) Gaseous emissions from automobiles.
- It is estimated that about 60% of the air pollution is caused by transportation vehicles, 20% comes from combustion processes for heating, power generation and incineration and 20% comes from industrial, commercial and other sources.

Nature of Air-pollutants

There are two types of air pollutants

- Particulate matter
- Gases

The gases are far more dangerous to humans.

(i) Particulate Matter. Particulates are small solid or liquid substances in the air resulting primarily from fuel combustion, incineration of waste materials, or industrial process losses.

Particulate matter includes smoke, soot, flyash, dusts, mists and fumes.

Individuals who breathe such substances into their lungs may suffer from respiratory diseases.

(ii) Sulphur oxides are formed by the combustion of fuel oil or coal; both are widely used to produce heat and power, Fog or thermal inversions can transform sulphur dioxide into toxic sulphur trioxide sulphuric acid mist, which can injure humans, animals, plant life and even damage buildings.

(iii) Ozone is a toxic gas that can cause irritation of eyes and the respiratory system.

(iv) Nitrogen oxides are toxic pollutants. They come from stationary combustion sources or from motor vehicle exhausts. They damage vegetation and corrode metals.

(v) Carbon monoxide is a notoriously deadly gas. It results from the incomplete combustion of carbon-containing materials, such as gasoline. In nonfatal quantities it can affect the eye-sight, produce nervous reactions,, cause headache, nausea etc.

(vi) Hydrocarbons are chemical compounds such as olefins, paraffins, naphthenes and aromatics. Hydrocarbons emit as a result of incomplete burning of gasoline in automobiles. Hydrocarbon pollution contributes to photochemical smog, lung and eye irritation and may inhibit vegetation growth.

(vii) Organic and inorganic acids are formed in the atomosphere as a result of combustion and various industrial processes. The organic acids include acetic, fumaric and tannic acids. The inorganic acids include sulphuric, hydrochloric hydrofluoric and hydrobromic acids.

Air Pollution Control

- Since unlike water, air is not self-purifying, the way the air can be cleansed is by cleaner air pushing the dirtier away or by particulate matter settling out from the air and contributing to ground pollution.
- For controlling air pollution, old plants/factories responsible for pollution will have to install air purifying equipment and new plants, even if they contribute minimally to air contamination, will have to locate in areas not subject to inversion cycles.
- Transport vehicles cause maximum amount of air pollution. The exhaust from cars, trucks, buses and aeroplanes is an example of this. Since the vehicles are mobile, the pollution control is more difficult.
- To obtain effective control over the discharge levels of these forms of transport, it is necessary to establish and enforce federal standards.
- In the case of nonmobile sources of pollution such as thermal power plants, industrial plants etc., pollutants may be prevented from entering the atmosphere by *electrostatic precipitators, scrubbers* or by some other means of removal.

Effects of Air Pollution

- Various effects of air pollution are :

(1) *Visibility reduction*

The first effect of air pollution is the reduction in visibility produced by the scattering of light from the surfaces of air-borne particles.

(2) *Material damage*

Direct damage to structural metals, surface coatings, fabrics and other materials is a frequent and widespread effect of air pollution.

(3) *Agricultural damage*

A large number of food, forage and crops have been shown to be damaged by air pollutants. The curtailed value results from leaf damage, stunting of growth, decreased size and yield of fruit, and destruction of flowers etc.

(4) *Physiological effects on Man amd Domestic Animals*

Air pollution can result in *bronchitis, lung carcinoma, opticirritation, changes in blood chemistry* etc.

(5) *Psychological effects*

Since fear is a recognizable element in public reactions to air pollution, the psychological aspects of the phenomenon cannot be ignored.

Source testing and monitoring air quality (or pollution)

- The basic difference between source testing and monitoring is that the former implies a relatively elaborate and complete set of measurements to establish a starting or final condition, whereas the latter implies simpler and less complete measurements at regular intervals of a property which increases or decreases in proportion to others.
- Monitoring of an effluent source involves the use of
  - *(i)* Smoke density meters
  - *(ii)* Closed circuit TV systems
  - *(iii)* Mirrors, which reflect the stack effluent to an observer in the plant.
  - *(iv)* Continuous recording analytical equipment.
  - *(v)* Equipment for periodic grab sampling and analysis over long periods of time.
- A simple and commonly used stack–monitoring device for particulate material consists of a source of light focused through the gas stream onto a photoelectric cell which in turn is wired to automatic recording equipment. As the stack loading increases, the light transmittance decreases and the indcx value increases.

Continuous Monitoring Devices for Fluc Gases

- Devices are available for analyzing sulphur and nitrous oxides, CO, H_2S, etc., in either ambient air or stack flue gas emissions.

These devices tend to be selective, analyzing only the portion of the gas intended. Systems include

(i) Gas Chromatography

(ii) Photometry

(iii) Fuel cell sensors

(iv) U.V. absorption

(v) Light scattering techniques etc.

Air Pollution Control Devices

Various air pollution control devices are

(i) Mechanical collectors

(ii) Electrostatic precipitators

(iii) Filters

(iv) Wet scrubbers

(v) Absorbers

(vi) Adsorbers.

(i) Mechanical Collectors use gravitational and /or centrifugal force to remove particulate matter.

- Where the air becomes laden with large particles, gravity provides a good means of separation. A gravity settling chamber is merely a *wide spot in the line* where the velocity of smoke or gas is reduced to allow particles to settle out.
- A Cyclone separator subjects the particles to forces several times that of gravity.

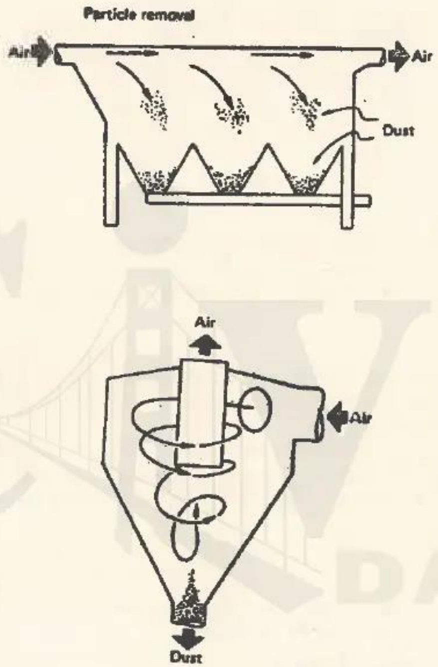

Fig. 33.1. (*a*) Gravity settling chamber (*b*) Cyclone separator.

(ii) Electrostatic Precipitators. An electrostatic precipitator is a means of putting an electric charge on the particles. The charge difference between the particle and the collecting plate causes attraction and collection.

(iii) Filters Consist of fibrous media, felted media and porous media. Filters are one of the most effective particle collection devices. The filter media stops the particles by impaction, interception diffusion settling and electrical attraction.

(iv) Wet Srcubbers

- Wet scrubbers can function as either or both particulate and gas control devices. They can handle hot gases and sticky particulates and liquids.
- Scrubbers remove particulate matter mainly by inertial impaction and they remove gases by absorption.

– Most wet scrubbers utilize atomized liquid drops as the collection targets for particle removal. Particles in the air are impinged against the water droplets.

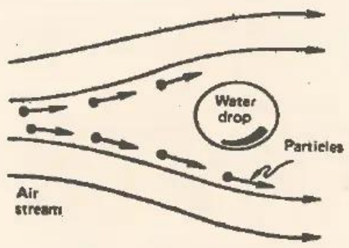

Fig. 33.2. Principle of Scrubber.

(v) Absorbers.

– Absorbers, as related to air pollution control, are a special category of wet scrubbers.

– Absorption towers work as effective pollution control devices. They utilize simple mechanical methods of achieving good contacting between the gas and the liquid phases to provide favourable overall mass transfer.

(vi) Adsorbers

– Adsorption process consists of contacting the solid with a gas mixture to remove any or all odor, taste, moisture, solvents or other pollutants from the gas. A few adsorbents are Activated carbon, Silica gel, Activated alumina etc.

– Adsorption is the taking up of a gas on the surface of a solid (or liquid). Physical adsorption consists of attracting gas molecules usually by electrostatic or van der Waals forces which result from gas molecule polarity and strongly positive or negative ions on the surface of the adsorbing solid.

– Adsorbers are usually constructed with the gases to be cleaned entering from the top and the cleaned gases leaving at the bottom. There are horizontal stationary adsorbent beds.

Dirty gases enter between the beds and adsorption occurs as part of the gases pass through the bed above and part pass through the bed below the gas inlet.

33.6. WATER POLLUTION AND CONTROL

Introduction

– Pollution is defined as something that adversely and unreasonably impairs the beneficial use of water. It includes addition of anything to water which changes the natural quality of water so that the downstream user does not receive the natural water of the stream.

In other words, water pollution is the specific impairment of water quality by agricultural, domestic or industrial wastes (including thermal and atomic wastes), to a degree that has an adverse effect upon any beneficial use of water yet that does not necessarily create an actual hazard to the public health.

– Water pollution may be defined as the addition to a natural body of water of any material which diminishes the optimal economic use of the water by the population which it serves.

– A waste is dumped into a body of water because a city, industry or individual wishes to eliminate a useless and somewhat noxious mess from its environment.

Classification of Water Pollutants

Water pollutants may be classified into following categories :-

(A) Chemical (B) Physical

(C) Physiological (D) Biological

(E) Radioactive

(A) Chemical pollutants

- Chemical materials discharged into a receiving water may be broadly classified as Organic, and Inorganic pollutants.
- Undesirable results from the discharge of inorganic materials include changes in the pH of the water caused by soluble salts and toxicity caused by heavy metals or other toxic materials.

Inert insoluble inorganics such as clay create sludge deposits on the bottom of the river and adversely affect the biological life.

- The major cosideration with respect to organic materials is the depletion of dissolved oxygen. Oils will form surface films, phenols will affect the taste and odor of water and refractory organics will cause death of fish and other aquatic life.

(B) Physical Pollutants

- Physical pollutants include

(*i*) Colour (*ii*) Turbidity

(*iii*) Temperature (*iv*) Suspended solids

(*v*) Foam (*vi*) Radio-activity.

- Colour though not necessarily harmful, is obviously undesirable in drinking water.
- Turbidity is primarily caused by either colloidal or very finely divided suspended matter which settles slowly and with difficulty. Turbidity caused by the hydrous oxides of Fe and Mn is quite objectionable in domestic water and may require special treatment for removal.
- Temperature increase (due to the use of water as cooling media in power generation) has only recently been considered as a water pollutant.
- Suspended materials may go in water from erosion processes or by various waste discharges.
- The resultant foam produced in many waters from detergents manufacturing industries may be objectionable from an aesthetic stand point.

(C) Physiological

- Undersirable taste and odor present in water used for drinking or food processing is objectionable to the consumer.
- Taste and odor of water can easily change if chlorophenols or H_2S is present in it, even in very small quantities.

(D) Biological

- It is most important of all water pollution problem. In fact, the single most important process in the water treatment plant is disinfection, which helps insure the absence of pathogenic organisms in the drinking water.
- Biological pollutants cause bacterial bone diseases, amoebic dysentery, cholera etc.

(E) Radioactive

- Radioactive pollution is the discharge of a radioactive waste material into a receiving body.

Sources of Water Pollution

Water, after being used for the following purposes, when drained out in a bigger and clean source of water, creates pollution problems over there

(*i*) Domestic life
(*ii*) Industry
(*iii*) Agriculture
(*iv*) Wild life watering
(*v*) Propagation of fish and other aquatic life
(*vi*) Swimming and bathing pools
(*vii*) Boating ponds/lakes
(*viii*) Water power generation
(*ix*) Transport etc.

Measurement of Water Pollution

– Waste water sampling and flow data are the two most important factors of any water pollution control program.

– Proper selection of sampling points assures samples that will yield representative and reliable data.

– Samples of water may be taken at plant outfalls, branch streams to main drain etc.

– Once the drain discharge and sample points have been established it is necessary to determine the type of sample to be collected at each point.

Samples are either *grab* or *composite*

A grab sample is a single sample, while the composite is one composed of a series of samples collected over a period of time and blended for analysis.

– Waste water streams can be continuously and automatically monitored and analyzed by reliable on-stream effluent monitoring equipment.

– A wide variety of instrumental methods for the chemical analysis of water constituents exist such as

(*i*) Atomic Absorption Spectrophotometry
(*ii*) Infrared Absorption Spectrophotometry
(*iii*) Electrochemical Analysis
(*iv*) Chemical Spectrophotometry
(*v*) Molecular Absorption Spectrophotometry.

Control of water pollution

– In order to maintain satisfactory water quality, effluents containing excessive amounts of objectionable materials must be treated to remove the offensive matter prior to the discharge of the effluent into a receiving water.

– In order to define the various processes available for water pollution control, it is necessary to categorically describe the process in generalized terms. The various phases of waste treatment are usually described as follows:-

(*i*) Pretreatment
(*ii*) Primary treatment
(*iii*) Secondary treatment
(*iv*) Tertiary treatment.

(i) Pretreatment

– It includes processes to remove larger aggregates of floating and suspended solid matter, grit and also much of the oil and grease content.

– It may also include flow measurement and sometimes prechlorination to prevent odors from emanating from the subsequent processes.

– The most common methods of removing large solid particles are passing the waste effluent through screens consisting of spaced metal bars or using comminutors to grind the solids into small pieces which subsequently settle out in the primary sedimentation tanks.

(ii) primary treatment

– It consists of the pretreatment processes plus tank sedimentation and usually chlorination prior to discharge into a receiving water.

– A primary treatment plant will often include the sludge digester, in which the solids removed from the sedimentation tank are subjected to anaerobic fermentation *i.e.,* stabilisation in the absence of free oxygen.

(iii) Secondary treatment

– It is a biological process following primary treatment.

– The most common biological process in use today is probably either the activated sludge process or an aerated lagoon.

– Activated sludge process consists of maintaining an active floc in a tank well supplied with oxygen in such a way that maximum contact is made between the incoming waste water and the organisms in the floc.

– Another secondary treatment process is *Trickling Filteration.* This process is identical to activated sludge process, except that the micro organisms working to stabilize the organic waste material are attached to a fixed bed rather than being of a floating and suspended nature. The waste water is distributed from rotary nozzles over the bed which gives support to the biological film. The material in the waste is oxidized after assimilation by the bacteria, and then released in a more stable form.

(iv) Tertiary treatment

– If the effluent from a secondary treatment plant is not considered satisfactory, tertiary treatment may be required.

– Tertiary treatment may consist of many different processes including

Coagulation	Membrane separation processes
Filtration	Adsorption etc.
Coprecipitation	

– *Coagulation* may be defined operationally as the reactions that take place upon the addition of a coagulant (*e.g.,* inorganic metal salts* and organic polymers) to water and result in the formation of insoluble products of reaction between the coagulant and the impurity to be removed.

– *Membranes* have been used to concentrate and separate soluble ions and molecules, colloidal species, particulates and micro organisms in waste water. The capabilities of membranes to exclude species by virtue of their size (dialysis and membrane filtration) and to adsorb or exclude due to specific chemical interactions (ion exchange and osmosis) have been utilized.

– In *Coprecipitation* an ion is removed from the solution phase with a precipitate (carrier), even though its solubility is not exceeded. The two principal mechanisms for this process are *Adsorption*, in which the ion coprecipitating is adsorbed onto the surface of the primary precipitate ; and *occlusion*, in which the ion is found in the interior of the primary precipitate, this category includes both solid solution formation and ion entrapment.

– The most common type of water treatment, except disinfection, is *filtration*. This was almost the only form of treatment practised prior to the advent of chlorination.

* e.g. aluminium Sulphate, Sodium aluminate etc.

Today, almost all water supplies are filtered as an integral part of the purification process. Filtration should be done after coagulation as most of the filters will not remove colloidal impurities by themselves.

Filters amay be made up of sand, anthracite coal or diatomaceous earth.

33.7. SOLID WASTE MANAGEMENT

Introduction

Solid waste management can be divided into two major areas

(*i*) *Collection* including storage, transfer and transport, and

(*ii*) *disposal*, including any accompanying treatment.

The *collection* operation can be sub-divided into two unit operations, *collection* and *haul*. The *collection* operation consists of removing solid waste from the storage point. The *haul* operation includes the total round trip travel time (for the vehicle) from the collection route to the (waste) disposal site.

Three alternatives are normally considered for solid waste disposal

(1) Direct shipment from municipalities to a sanitary landfill.

(2) Direct shipment from municipalities to a transfer station where solid waste is transferred to larger vehicles and then shipped for ultimate disposal.

(3) Direct shipment from municipalities to an incinerator where the solid waste is burned and the residue is shipped for ultimate disposal.

Solid waste management planning requires an assessment of many complex interactions among transportation systems, land use patterns, urban growth and development, and public health considerations.

Waste handling methods

The physical nature of a waste may determine its handling characteristics. The following are common handling methods:——

(*i*) Solids,semi-solids, some wet materials, sticky, or tarry substances may be handled by front-end loaders or buckets.

(*ii*) Viscous liquids may be pumped by special pumps.

(*iii*) Liquids are handled by normal pumping equipment.

(*iv*) Packages may be handled in cartons, and

(*v*) Some materials are handled in fiber-pack drums.

Solid waste shredders

– Such machines can convert rubbish into a form more easily and economically handled for processing. Hammer mills in one adaptation or another, are the most commonly used size reduction machines.

Compactors

Such machines can compact refuse. Hydraulic or pneumatic cylinders exert forces as high as 50 tons and reduce the original volume of refuse by 60 to 80%.

Incineration equipment

Solid and chemical wastes are often disposed of with the help of incinerators. Incinerators can be classified on the basis of burner chamber or fuel bed.

Common practices for solid waste disposal

– In most solid waste disposal methods, the main aim is to treat the refuse in such a way as to render it safe and sterile so that upon returning it to the environment, it will not pollute the air, water or land.

– There are presently only three disposal methods which are practical for most industrial applications:–

(1) *Haulaway loose*. Waste is removed from plant for disposal by means such as landfill, incineration etc. Hauling away means exporting the refuse as loose material.

(2) *Haulaway Compacted*. If the amount of refuse is large, refuse is compacted and then disposed off. It is a very popular waste disposal method for industrial installations.

(3) *On site incineration* is generally accepted as a good method for disposing of solid wastes. Industrial waste is burned efficiently and economically without polluting the air.

Solid Waste Salvage and Recovery

Many experts feel that the only lasting solution for the solid waste disposal problem lies in recycling and reuse of wastes (Fig. 33.3).

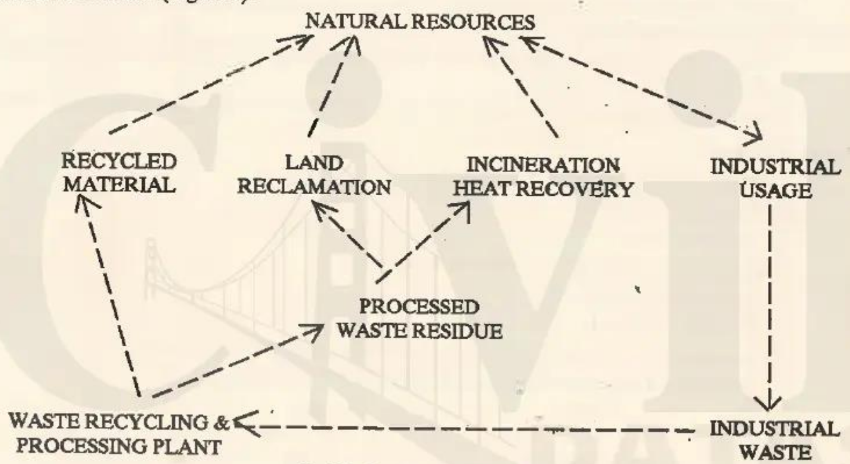

Fig. 33.3. Recycling industrial waste.

Waste processing has the following advantages:–

(*i*) Added revenue

(*ii*) Less waste to be disposed of.

(*iii*) Less transportation costs of waste.

(*iv*) Processed residue waste is put into a form which makes it suitable for land reclamation.

– *Steps involved in salvaging and recovery*

(*i*) Receiving the raw industrial waste and conveying it to a salvage-separation area.

(*ii*) Separating the salvageable materials from waste materials to be further processed.

Ferrous and nonferrous materials can be separated using magnets, cardboard and paper products can also be removed and placed onto the paper salvage conveyor for entry into the paper processing system

(*iii*) Unsalvaged waste residue is conveyed to the main pulverizer unit,

(*iv*) Compaction system is employed to compact the pulverized residue for disposal.

33.8. NOISE AND ITS CONTROL

Introduction

- Noise may be defined as *unwanted sound.*
- Sound is caused by any disturbance in the air. Sound travels in air at a speed of approximately 350 m per second. When the pressure waves meet human ear they cause a sensation which we also call sound.
- Noise is a form of air pollution and like other forms of pollution, it affects the quality of life and so can be thought of as a social cost.
- The noise producers bear only a part of this cost and for them this is outweighed easily by the benefit they get from their industry.

The major part of this burden falls upon people who live around or are present there and they are not directly party to any benefits.

- Noise annoys, distracts, disturbs and when exposure to it is sufficient, noise can cause physiological effect leading to deafness.

Noise causes disturbance with consequent reduction in productivity, efficiency, accuracy and safety.

Sources of Noise

- Sources of noise may be classified as follows:–

(*i*) *Outdoor sources* such as

road traffic noise

noise around airports [peak level over 110 dB(A)]

construction site noise, and

noise from industries.

(*ii*) *Indoor source* implies noise from within the factory and other places of work.

- Given below are the noise levels produced from different sources of noise.

Noise	dB(A) (Decibel)
Large jet airliner	140
Riveting on steel plate	130
Train on steel bridge	110
Weaving shed	100
Heavy road traffic	85
Light road traffic	55

Noise Control

(*a*) *Method of measuring noise level*

Sound level meters are placed around the source to measure the noise level from different noise sources.

(*b*) *Approach*

- Once the noise has been established in the air, there is almost no way in which it can be controlled by means of a power driven device or an electrical *noise level controller.*
- Noise control can be achieved mainly by planning and forethought.
- As long as there is some method of controlling the noise, the method of approach can be analysed into

(*i*) Control (of noise) at the source.
(*ii*) Control along the path of sound between the source and receiver (listener or victim) and
(*iii*) at the receiver.

Noise Control at the source

- The control of noise at the source is usually the cheapest and most reliable method of noise control.
- Noise from a particular source can often be reduced by means of an *enclosure. Sound absorbing material* should be included within the enclosure.
- To control noise, the impacting parts of the machines should be enclosed.
- All rotating or impacting machines should be based on antivibration mountings.
- All rigid connections of the machine in the way of electricity, water, air, gas, etc., should have vibration decouplers around the machine.
- Internal combustion engines should be properly silenced.

Noise control along path of sound.

- Once the sound has left its source, it must travel some distance before reaching the place at which the noise nuisance will occur. Common paths for noise are airborne paths such as ducts and corridors and it is usual to include here walls that break up airborne sound paths.
- When a noise source is directly coupled to conducting paths such as pipes or air ducts, these paths can carry sound energy. To control noise along the path, one need to consider how sound is transmitted through ducts, corridors, directly through walls and along pipes and through the structure of buildings. In such cases contacting the manufacturers of fans and ducting system can be of much help. However, the following points may also be kept in mind.

(*i*) An effective method of increasing the sound insulation of a wall is to construct the wall of a number of separate layers.

(*ii*) Sound absorbing material has a definite role to play in improving sound transmission loss of a panel.

- To control noise along the path in a factory the following, may be employed :

(*i*) *Noise barriers*, which involve frequency dependent diffraction effects which depend on the size and geometry of the barrier. Noise barriers do not eliminate sound energy, but merely reflect and/or diffract it somewhere else.

(*ii*) *Enclosures* having inside lined with sound absorbing material such as fibre glass.

(*iii*) *Noise refuges* which are nearby rooms where the factory produced noise level does not exceed 85 dB (A). Employees can use these rooms for off-time activities, paperwork etc.

(*iv*) *Noise screens* limit the spread of noise and provide quiet area at places of sedentary work. Noise screens are used where noise refuges cannot be provided such as in large areas where many processes are occurring.

Noise control at the receiver

- Noise reaches the people and the delicate instruments. The noise level therefore is controlled by treating the area within which the receiver is located. For this, one has to study the acoustics of the situation.
- The use of sound absorbers is very effective. Soft furnishings, carpets, human beings, windows, resonant panels and so on all contribute to the sound absorption and so does the air within the room.